

Deepak Singh

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EDUCATION

- **Graphic Era Hill University**, Dehradun Sept 2023 – Jun 2027
B.Tech in Computer Science (AI/ML) — CGPA: 7.89/10.0
- **Whitehall Sr. Secondary School**, Uttarakhand 2020 – 2023
Class 12th (PCM): 82.7% — Class 10th: 88.2%

PROJECTS

- **DocuMind – RAG-based Document Q&A System** — 🐙 Jul – Aug 2025
Python, LangChain, FAISS, HuggingFace Transformers, FastAPI, Streamlit
 - Built a Retrieval-Augmented Generation (RAG) pipeline for natural language Q&A over PDFs using LangChain and FAISS vector store
 - Integrated HuggingFace sentence-transformers for semantic embeddings achieving 87% answer relevance on 200-query eval set
 - Deployed end-to-end via FastAPI backend and Streamlit UI supporting multi-document ingestion with chunk overlap tuning
- **Rain Prediction using Machine Learning** — 🐙 Nov 2024 – Jan 2025
Python, Scikit-learn, XGBoost, Pandas, NumPy
 - Built end-to-end ML pipeline processing 145K+ weather records achieving 82% accuracy and 0.91 ROC-AUC
 - Engineered 5+ meteorological features (temp/humidity differentials) improving F1-score by 12% through feature engineering
 - Optimized XGBoost with GridSearchCV across 50+ hyperparameter combinations, outperforming baseline by 15%
- **SentimentAPI – Fine-tuned BERT Classifier** — 🐙 Mar – Apr 2025
Python, HuggingFace Transformers, PyTorch, FastAPI, Docker
 - Fine-tuned DistilBERT on 50K IMDB reviews reaching 93.4% accuracy vs 89.1% baseline
 - Exported to ONNX reducing inference latency by 38%; containerized with Docker and served via FastAPI at 200+ req/sec
- **JavaFX Algorithm Visualizer** — 🐙 Apr – Jun 2025
Java, JavaFX, MVC Architecture, Scene Builder
 - Developed interactive visualization platform for 5 sorting algorithms with real-time animation rendering at 60 FPS
 - Integrated performance benchmarking comparing execution time and Big-O complexity across 1,000 elements

TECHNICAL SKILLS

- **Languages:** C++ (STL, OOP), Python (NumPy, Pandas), Java (Collections, JavaFX), SQL, HTML/CSS
- **ML & AI:** Scikit-learn, PyTorch, TensorFlow, Keras, XGBoost, HuggingFace Transformers, LangChain, FAISS, Feature Engineering, Hyperparameter Tuning, RAG, Fine-tuning
- **GenAI & LLMs:** Prompt Engineering, Embedding Models, Vector Databases (FAISS, ChromaDB), HuggingFace Hub
- **Model Evaluation:** ROC-AUC, Precision, Recall, F1-score, Cross-Validation, ONNX Quantization
- **Tools & Databases:** Git/GitHub, FastAPI, Streamlit, Docker (basics), MySQL, MongoDB, Jupyter, VS Code
- **Core CS Fundamentals:** Data Structures & Algorithms, Dynamic Programming, Graph Algorithms, DBMS, OOP, OS, Computer Networks

ACHIEVEMENTS & CERTIFICATIONS

- Solved 640+ DSA problems on LeetCode with contest rating of 1618 focusing on DP, Graphs, Trees, and Binary Search
- Google Cloud Computing Foundations (NPTEL) – Scored 90% covering GCP, Compute Engine, BigQuery, IAM (Nov 2024)
- Completed 10+ hands-on Google Cloud labs in ML APIs, cloud infrastructure, and data analytics
- Salesforce Trailhead Virtual Internship – 2-month training in CRM automation and Lightning platform (July 2025)